1 At least 3 terms correct from $6x^2 + 15x - 4x - 10$ **[M1]**

$6x^2 + 11x - 10$ **[A1]**

[Total Marks: 2]

2 (a) Expansion to $6x - 3 - 2x - 8$ **[M1]**

$4x - 11$ **[A1]**

(b) Extracting a partial common factor, e.g. $5xy(3x - 4y)$ or $x(15xy - 20y^2)$ or $5(3x^2y - 4xy^2)$ **[M1]**

$5xy(3x - 4y)$ **[A1]**

[Total Marks: 4]

3 (a) M1 for $8a - 12b$ or $-3a + 15b$ seen

$5a + 3b$ **[A1]**

(b) M1 for a correct partial factorisation, e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$

$6xy(2x - 3y)$ **[A1]**

[Total Marks: 4]

4 (a) $8x - 12y - 3x + 15y$ **[M1]**

$5x + 3y$ **[A1]**

(b) $7ab$ extracted as a common factor **[M1]**

$7ab(2a - 3b)$ **[A1]**

[Total Marks: 4]

5 (a) $12x - 8 - 2x + 10$ **[M1]**

$10x + 2$ **[A1]**

(b) $5ab(3a - 2b)$ **[B2]**

If not B2, then B1 for $5a(3ab - 2b^2)$ or $5b(3a^2 - 2ab)$ or $ab(15a - 10b)$ **[B1]**

[Total Marks: 4]

6 (a) $6x - 15 - 2x - 8$ **[M1]**

$4x - 23$ **[A1]**

(b) $4xy(2x - 3y)$ or $4x(2xy - 3y^2)$ or $4y(2x^2 - 3xy)$ or $xy(8x - 12y)$ **[M1]**

$4xy(2x - 3y)$ **[A1]**

[Total Marks: 4]

7 **(a)** Correct expansion: $6x - 15 - 4x + 8$ **[M1]**

$2x - 7$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

Correct partial factorisation e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$ **[M1]**

[Total Marks: 4]

8 **(a)** $5a$ or $+3b$ **[B1]**

$5a + 3b$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

If B0, allow [M1] for a correct partial factorisation, e.g. $6x(2xy - 3y^2)$ or $3xy(4x - 6y)$

[Total Marks: 4]

9 **(a)** Expansion to give $6x - 15$ and $-4x + 12$ **[M1]**

$2x - 3$ **[A1]**

(b) $7xy(2x + 3y)$ **[B2]**

Partial factorisation, e.g. $7x(2xy + 3y^2)$ or $xy(14x + 21y)$ **[B1]**

[Total Marks: 4]

10 **(a)** $3(4x^2 - 9y^2)$ or $(6x - 9y)(2x + 3y)$ or $(2x - 3y)(6x + 9y)$ **[M1]**

$3(2x - 3y)(2x + 3y)$ **[A1]**

(b) $2a(x - 3y) + b(x - 3y)$ or $x(2a + b) - 3y(2a + b)$ **[M1]**

$(2a + b)(x - 3y)$ **[A1]**

[Total Marks: 4]

11 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $(y - 4x)(y + 4x)$ **[B1]**

(c) $(p + q)(p - 5q)$ **[B2]**

If B0, allow [M1] for $(p + aq)(p + bq)$ where $ab = -5$ or $a + b = -4$

[Total Marks: 5]

12 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $c(a - 3b) + 2d(a - 3b)$ or $a(c + 2d) - 3b(c + 2d)$ **[M1]**

$(a-3b)(c+2d)$ **[A1]**

(c) $(4x-5y)(4x+5y)$ **[B1]**

[Total Marks: 5]

13 (a) M1 for 3 correct terms out of $2x^2+8x-3x-12$

$2x^2+5x-12$ **[A1]**

(b) M1 for $x(a-b)$ or $-3y(a-b)$ seen

$(x-3y)(a-b)$ **[A1]**

(c) $(7x-4y)(7x+4y)$ **[B2]**

If 0 scored, allow M1 for $(7x-4y)(7x-4y)$ or $(7x+4y)(7x+4y)$

[Total Marks: 6]

14 (a) $(x+a)(x+b)$ where $ab=15$ or $a+b=-8$ **[M1]**

$(x-3)(x-5)$ **[A1]**

(b) $(2x+a)(x+b)$ where $ab=-12$ or $2b+a=-5$ **[M1]**

$(2x+3)(x-4)$ **[A1]**

(c) $a(x-3)+2b(x-3)$ or $x(a+2b)-3(a+2b)$ **[M1]**

$(a+2b)(x-3)$ **[A1]**

[Total Marks: 6]

15 (a) $6x^2+2x-15x-5$ (at least 3 correct terms) **[M1]**

$6x^2-13x-5$ **[A1]**

(b) $(x+a)(x+b)$ where $ab=-18$ or $a+b=-7$ **[M1]**

$(x-9)(x+2)$ **[A1]**

(c) $2(9x^2-25y^2)$ **[M1]**

$(3x-5y)(3x+5y)$ seen **[M1]**

$2(3x-5y)(3x+5y)$ **[A1]**

[Total Marks: 7]

16 (a) Correct expansion of any two brackets, e.g. $2x^2-x-6$ or $2x^2-7x-15$ or $x^2-7x+10$ **[M1]**

Correctly multiplies their 3-term quadratic by the remaining bracket to get at least 4 correct terms **[M1]**

$2x^3-11x^2-x+30$ **[A1]**

(b) Extracts common factor x to give $x(12x^2+17x-5)$ **[M1]**

Correctly factorises the quadratic expression to $(4x-1)(3x+5)$ **[M1]**

$x(4x-1)(3x+5)$ **[A1]**

[Total Marks: 6]

17 **(a)** Taking out common factor to reach $2(9x^2-25y^2)$ **[M1]**

Factorising difference of squares to reach $(3x-5y)(3x+5y)$ **[M1]**

$2(3x-5y)(3x+5y)$ **[A1]**

(b) Taking out common factor x to reach $x(3x^2+10x-8)$ **[M1]**

Factorising quadratic to reach $(3x-2)(x+4)$ **[M1]**

$x(3x-2)(x+4)$ **[A1]**

[Total Marks: 6]

18 **(a)** Correct expansion of two brackets, e.g. $(x^2+x-6)(2x-1)$ or $(x-2)(2x^2+5x-3)$ **[M1]**

Multiplication of quadratic by third bracket yielding 6 terms, e.g. $2x^3-x^2+2x^2-x-12x+6$ **[M1]**

$2x^3+x^2-13x+6$ **[A1]**

(b) **(i)** $(3x+a)(2x+b)$ where $ab=-10$ or $2a+3b=-11$ **[M1]**

$(3x+2)(2x-5)$ **[A1]**

(ii) $a(x-y)+3c(x-y)$ or $x(a+3c)-y(a+3c)$ **[M1]**

$(a+3c)(x-y)$ **[A1]**

[Total Marks: 7]

19 **(a)** $x(c-5d)+3y(c-5d)$ or $c(x+3y)-5d(x+3y)$ **[M1]**

$(x+3y)(c-5d)$ **[A1]**

(b) $2x(x^2-9)$ **[M1]**

$2x(x-3)(x+3)$ **[A1]**

(c) $2x(6x^2-7x-5)$ **[M1]**

$2x(3x-5)(2x+1)$ **[A2]**

If not A2, then A1 for $(3x-5)(2x+1)$ or $2x(3x+5)(2x-1)$ **[A1]**

[Total Marks: 7]

20 **(a)** Expands any two brackets, e.g. $(2x^2-7x-4)(x+3)$ **[M1]**

Multiplies their quadratic by the third bracket, e.g. $2x^3-7x^2-4x+6x^2-21x-12$ **[M1]**

$2x^3-x^2-25x-12$ **[A1]**

(b) $2a(x+y)-3b(y+x)$ or $x(2a-3b)+y(2a-3b)$ **[M1]**

$(x+y)(2a-3b)$ **[A1]**

(c) $(2x+5)(3x-4)$ **[B2]**

If B0, allow [M1] for $(2x+a)(3x+b)$ where $ab=-20$ or $3a+2b=7$

[Total Marks: 7]

21 (a) $(2a-b)(3x-y)$ **[B2]**

Correct partial factorisation e.g., $2a(3x-y)-b(3x-y)$ or $3x(2a-b)-y(2a-b)$ **[M1]**

(b) (i) Extracts common factor of 2: $2(25x^2-4y^2)$ **[M1]**

Factorises difference of two squares: $(5x-2y)(5x+2y)$ **[M1]**

$2(5x-2y)(5x+2y)$ **[A1]**

(ii) Extracts common factor of x: $x(2x^2+5x-12)$ **[M1]**

Factorises quadratic: $(2x-3)(x+4)$ **[M1]**

$x(2x-3)(x+4)$ **[A1]**

[Total Marks: 8]

22 (a) Correct expansion of any two brackets, e.g., $(2x^2-5x-3)$ or $(2x^2+9x+4)$ or (x^2+x-12) **[M1]**

Correct expansion of their quadratic with the third bracket, producing at least 5 terms, e.g., $2x^3+8x^2-5x^2-20x-3x-12$ **[M1]**

$2x^3+3x^2-23x-12$ **[A1]**

(b) $(2x+3)(x-5)$ **[B2]**

Accept $(2x+a)(x+b)$ where $ab=-15$ or $2b+a=-7$ **[M1]**

[Total Marks: 5]

23 (a) Four terms with at least three correct from $2x^2+10x-3x-15$ **[M1]**

$2x^2+7x-15$ **[A1]**

(b) $(x+a)(x+b)$ where $ab=-18$ or $a+b=-7$ **[M1]**

$(x-9)(x+2)$ **[A1]**

(c) $(2x+a)(2x-a)$ or $(cx+5y)(cx-5y)$ **[M1]**

$(2x-5y)(2x+5y)$ **[A1]**

[Total Marks: 6]

24 (a) $(x-4)(2x^2+7x+6)$ or $(2x^2-5x-12)(x+2)$ or $(x^2-2x-8)(2x+3)$ **[M1]**

$2x^3+7x^2+6x-8x^2-28x-24$ (allow one error in expansion) **[M1]**

$2x^3-x^2-22x-24$ **[A1]**

(b) $(4x - 7y)(4x + 7y)$ **[B2]**

If not B2, then B1 for $(4x - ky)(4x + ky)$ or $(kx - 7y)(kx + 7y)$ where $k \neq 0$ **[B1]**

(c) $(3x + 2)(x - 4)$ **[B2]**

If not B2, then B1 for $(3x + a)(x + b)$ where $ab = -8$ or $3b + a = -10$ **[B1]**

[Total Marks: 7]

25 (a) Expands any two brackets correctly, e.g., $(2x^2 - 5x - 12)(x + 1)$ or $(x - 4)(2x^2 + 5x + 3)$ or $(x^2 - 3x - 4)(2x + 3)$ **[M1]**

Multiplies their resulting quadratic by the third bracket, allowing one error **[M1]**

$2x^3 - 3x^2 - 17x - 12$ **[A1]**

(b) $(3x + a)(2x + b)$ or $(6x + c)(x + d)$ where $ab = -10$ or $cd = -10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $x^2(x + 3) - 4(x + 3)$ or $x(x^2 - 4) + 3(x^2 - 4)$ **[M1]**

$(x^2 - 4)(x + 3)$ **[M1]**

$(x - 2)(x + 2)(x + 3)$ **[A1]**

[Total Marks: 8]

26 (a) M1 for correctly expanding any pair of brackets, e.g., $2x^2 - 3x - 2$ or $2x^2 + 7x + 3$ or $x^2 + x - 6$

M1 for multiplying their quadratic expression by the third bracket, allowing one error in terms

$2x^3 + 3x^2 - 11x - 6$ **[A1]**

(b) M1 for $(3x + a)(2x + b)$ where $ab = -10$ or $3b + 2a = -11$

$(3x + 2)(2x - 5)$ **[A1]**

(c) M1 for partial factorisation extracting x: $x(3x^2 + 5x - 2)$

M1 for factorising the quadratic factor: $(3x - 1)(x + 2)$

$x(3x - 1)(x + 2)$ **[A1]**

[Total Marks: 8]

27 (a) $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ or $(x + 3)(2x^2 - 5x + 2)$ **[M1]**

$2x^3 - x^2 + 2x^2 - x - 12x + 6$ (allow one error in expansion) **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) $(3x \pm a)(2x \pm b)$ where $ab = 10$ or $6x^2 - 15x + 4x - 10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $2x(x^2 - 9)$ **[M1]**

$(x - 3)(x + 3)$ seen **[M1]**

$2x(x-3)(x+3)$ **[A1]**

[Total Marks: 8]